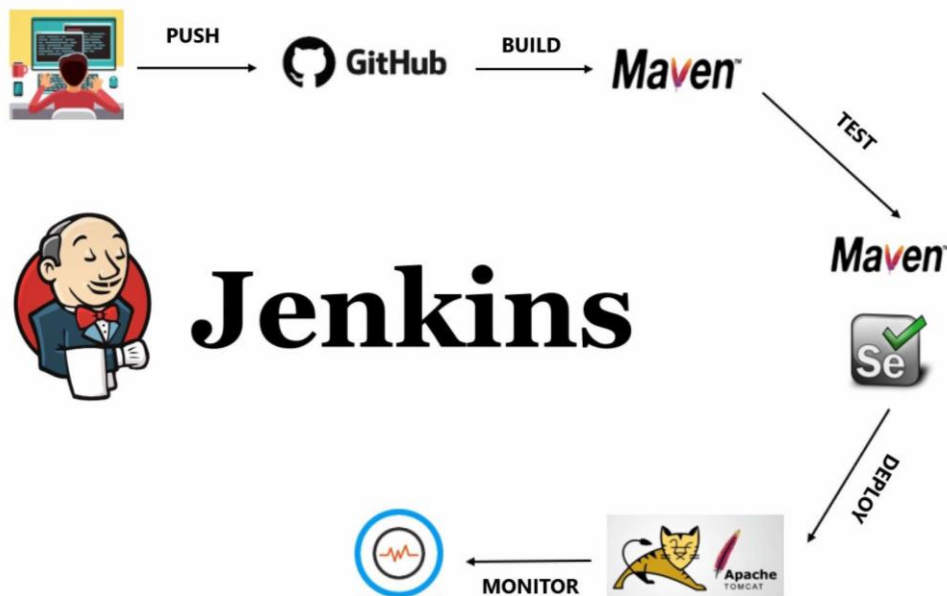




Jenkins

What is Jenkins?

- Jenkins is a free and open-source automation tool.
- It helps developers build, test, and deploy code automatically.
- It is written in Java and runs on the Java platform.
- Jenkins supports continuous integration and end-to-end automation of projects.



Core Concepts :-

The following are some important concepts of Jenkins:

1. Jenkins Controller (Master)

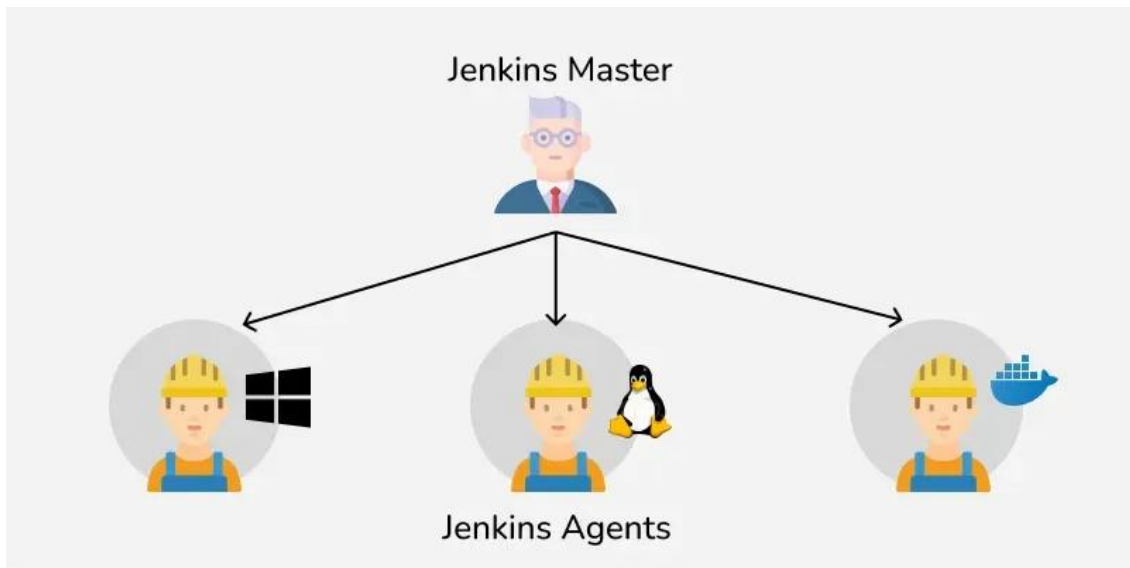
- Central system of Jenkins.
- Manages agents and assigns tasks.
- Loads plugins and controls job execution order.

2. Jenkins Agent (Slave)

- A machine that runs tasks like build, test, deploy.
- Controlled by the Jenkins Controller.
- Can have different OS, software, and hardware.

Types of Agents:

1. **Permanent Agents** → Always connected and ready.
2. **Ephemeral Agents** → Created only when needed and removed after work.



3. Jenkins Node

- Any machine where Jenkins jobs can run (Controller or Agent).
- Jenkins monitors node health and disables if issues occur.

4. Jenkins Job

- An automated task for building, testing, or deploying code.
- Saves time and reduces manual errors.

5. Jenkins Plugins

- 2000+ plugins available for integration and automation.
- Connects easily with tools like GitHub, Slack, Docker, AWS.
- Install and update plugins from the Dashboard.

6. Jenkins Pipeline

- Automates the CI/CD workflow.
- Improves productivity, speed, and application quality.

Jenkins Features :-

1. **Easy Setup:** Works on Windows, Linux, and macOS with a simple web-based configuration.
2. **Free & Open-Source:** Anyone can download, use, and customize Jenkins.
3. **Large Plugin Support:** 2000+ plugins available to integrate with almost any tool.
4. **CI/CD Automation:** Automatically builds, tests, and deploys code after updates.
5. **Strong Security:** Provides MFA login, role-based access, and secure server/user management.c

How to Install Java on Windows ?

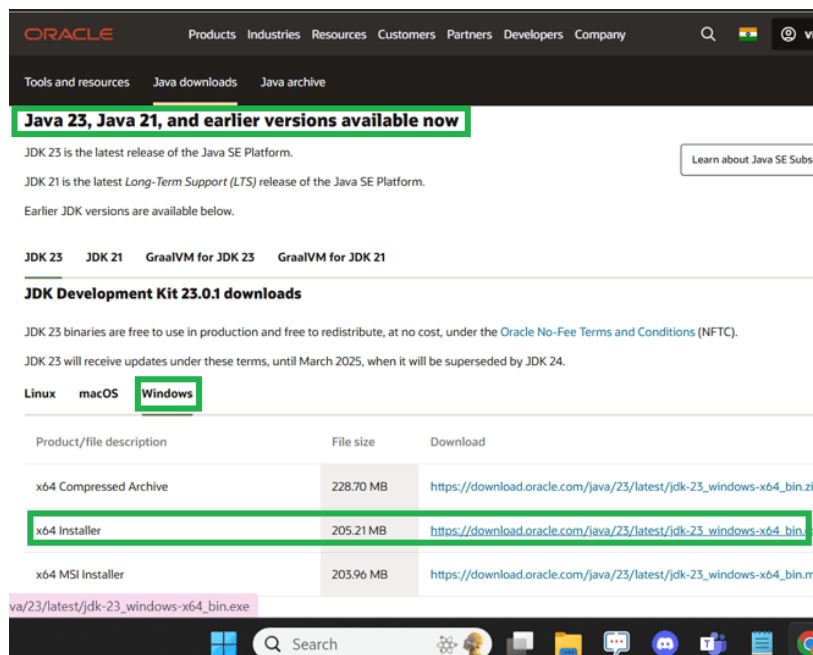
- Java is a versatile and widely used programming language.
- It is used to build applications for desktop, web, and mobile.
- To start coding in Java, you must install the Java Development Kit (JDK).

Steps to Download and Installing Java on Windows :-

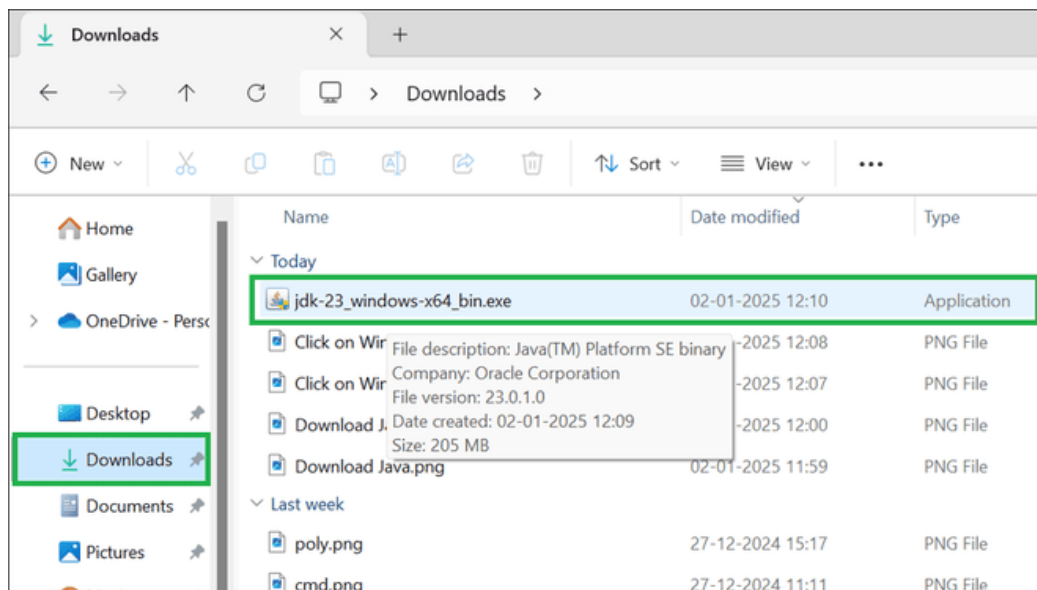
Step 1: Visit Oracle's website and click Download Java.



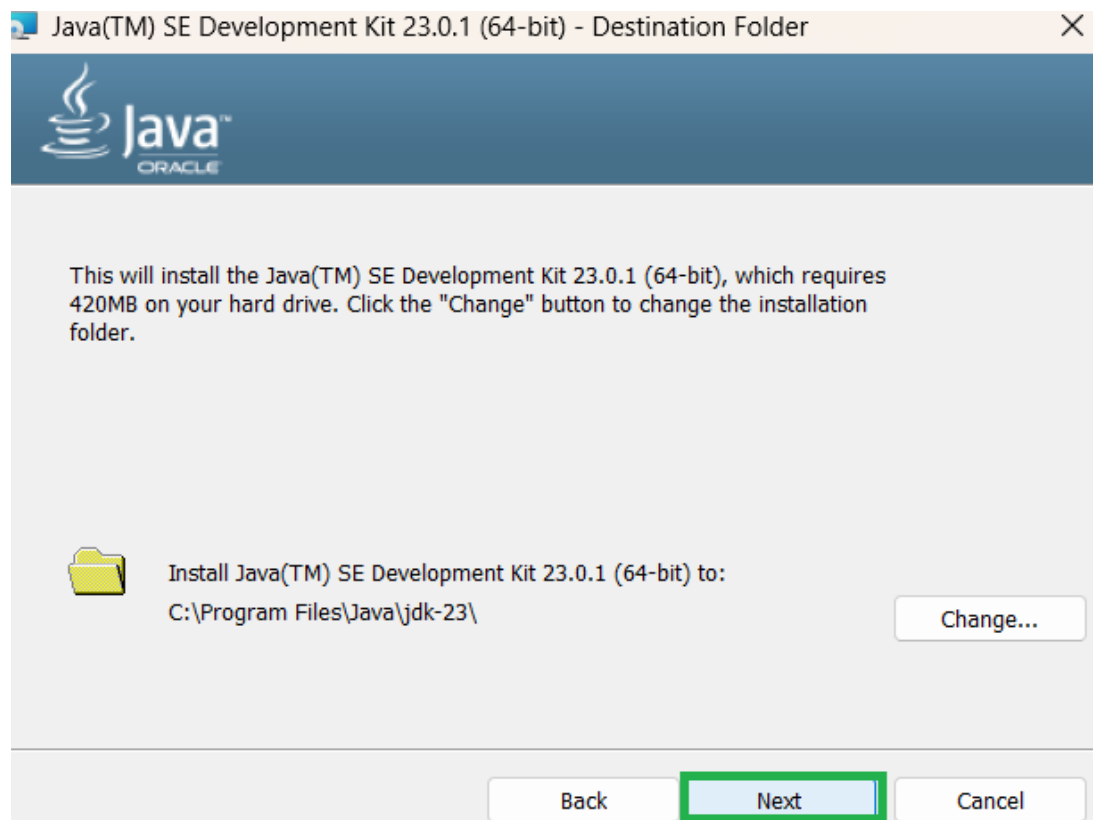
Step 2: Select Windows → x64 Installer to download the JDK 23 .exe file.



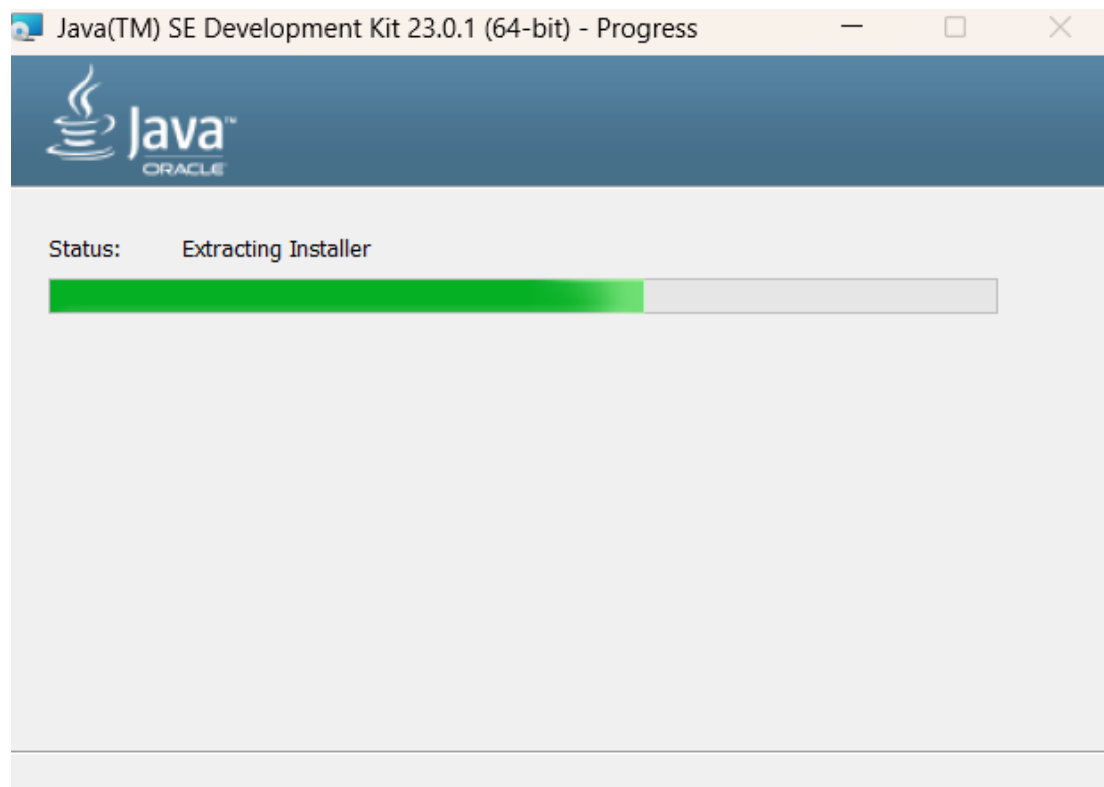
Step 3: Open the downloaded jdk-23_windows-x64_bin.exe file and click Next in the Java Installation Wizard.



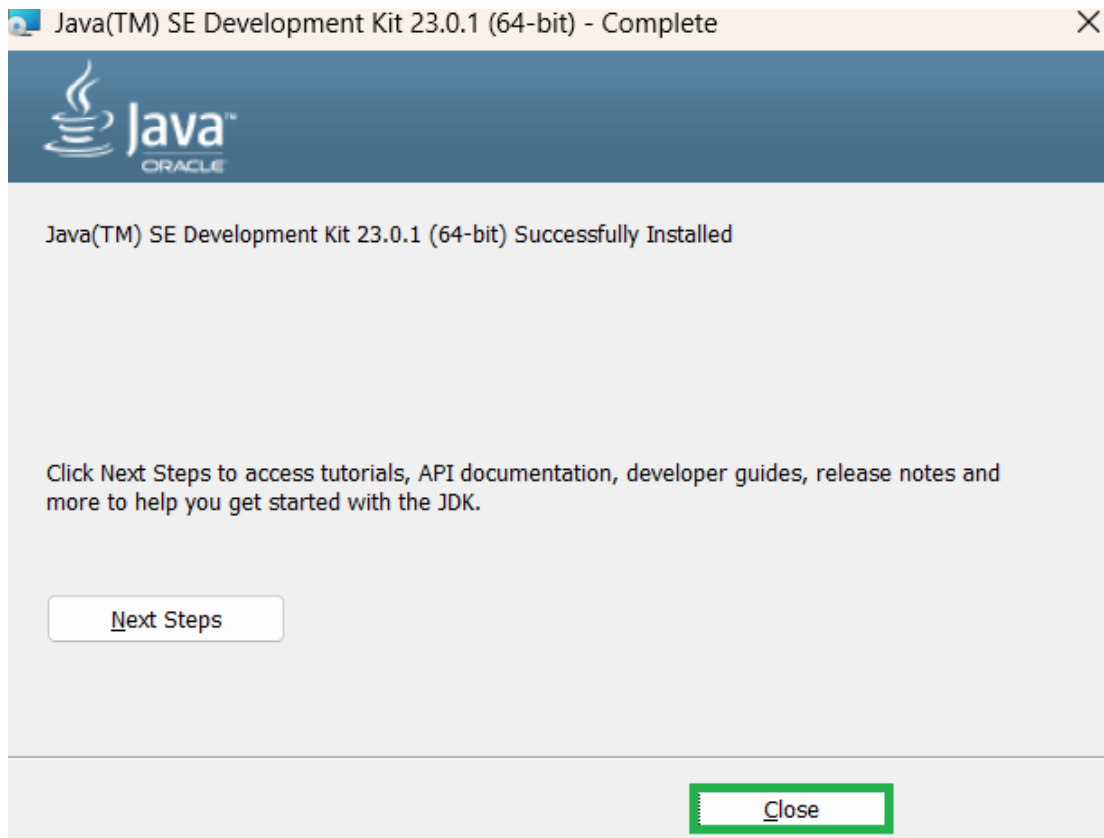
Again click on the **Next** button to install JDK.



The installation will begin as shown below:



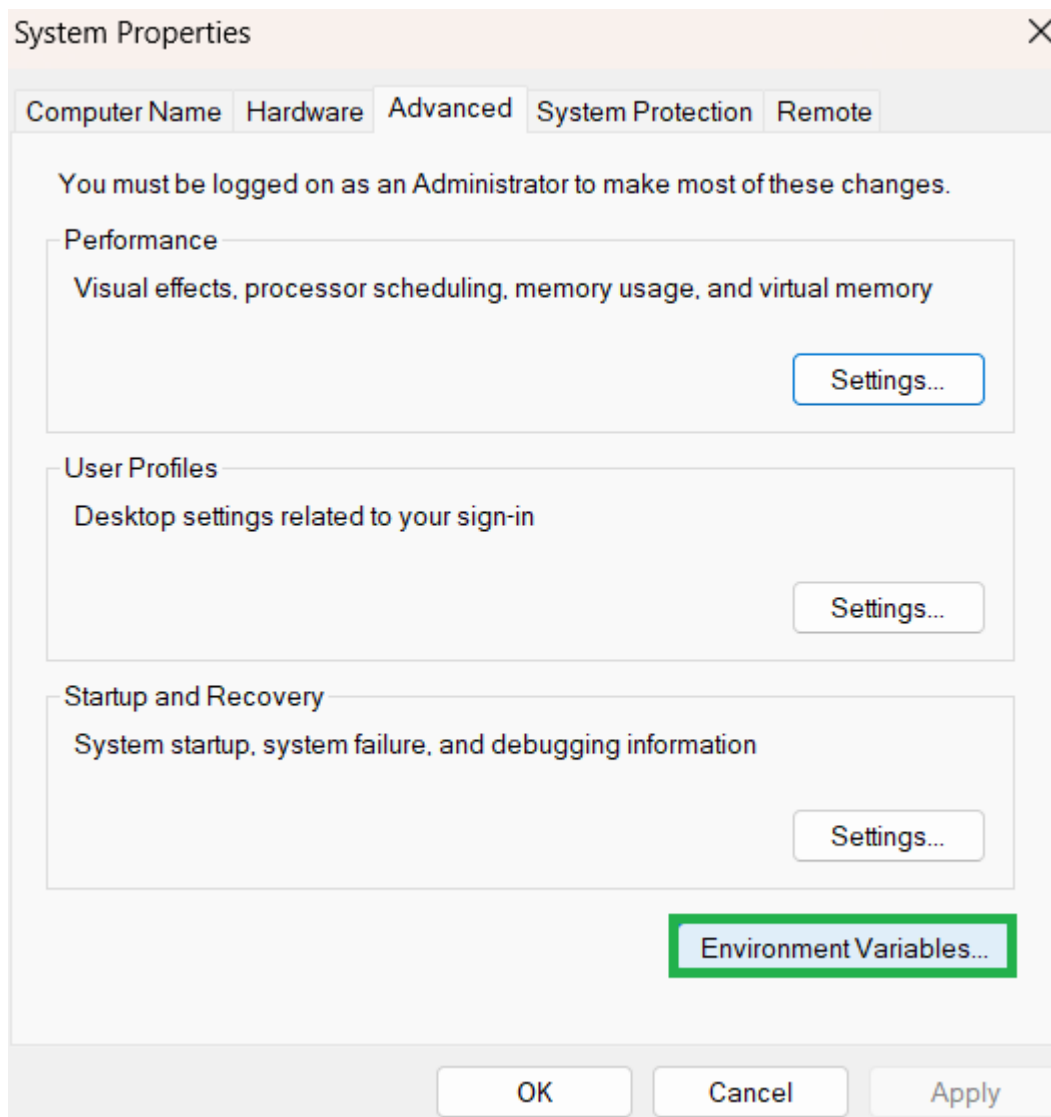
The Java installation is successfully completed as shown below. Now click on "**close**" to finish.



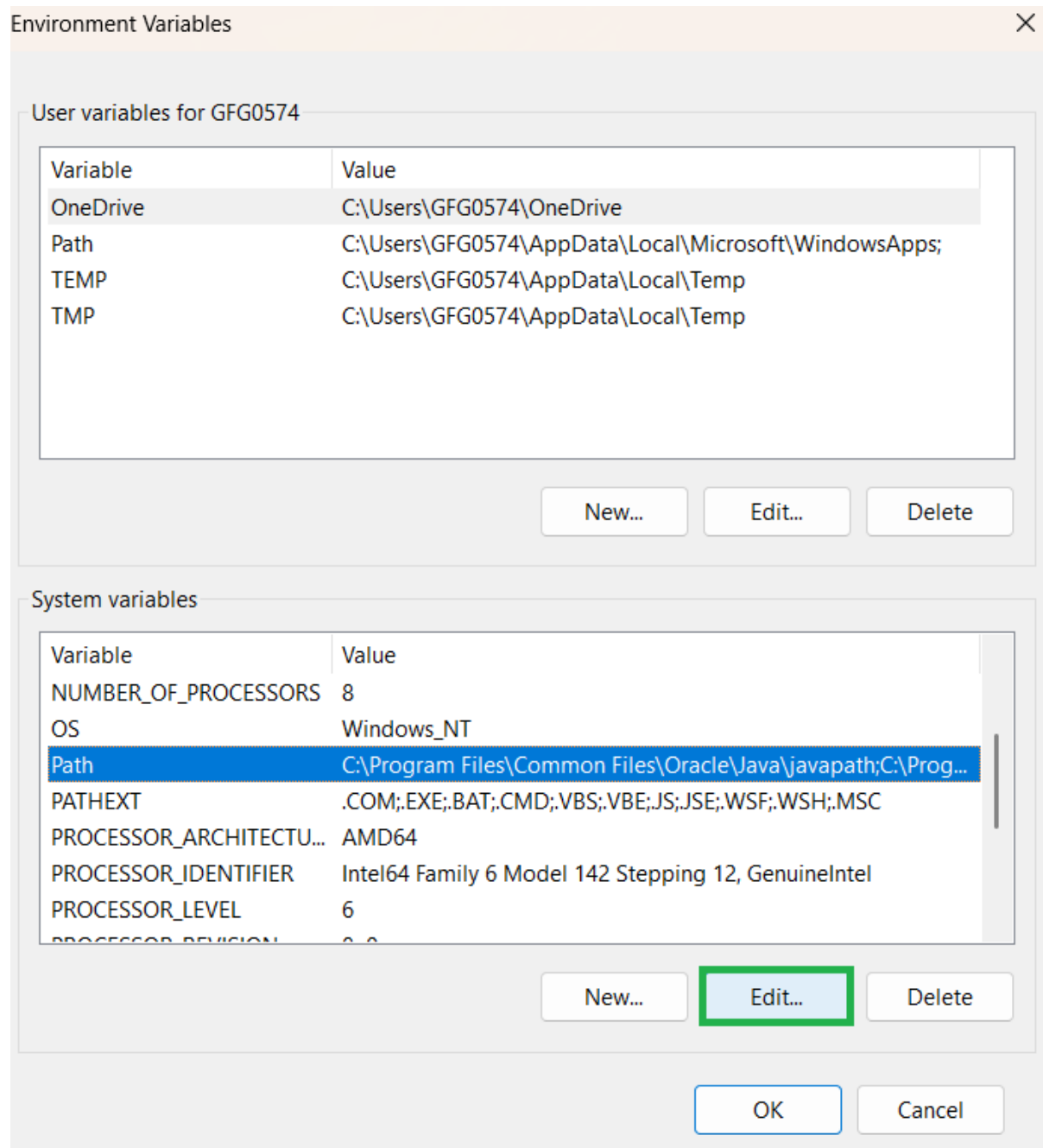
Step 4: Copy the path of the bin folder in C:\Program Files\Java\jdk-23 to set up Windows environment variables.



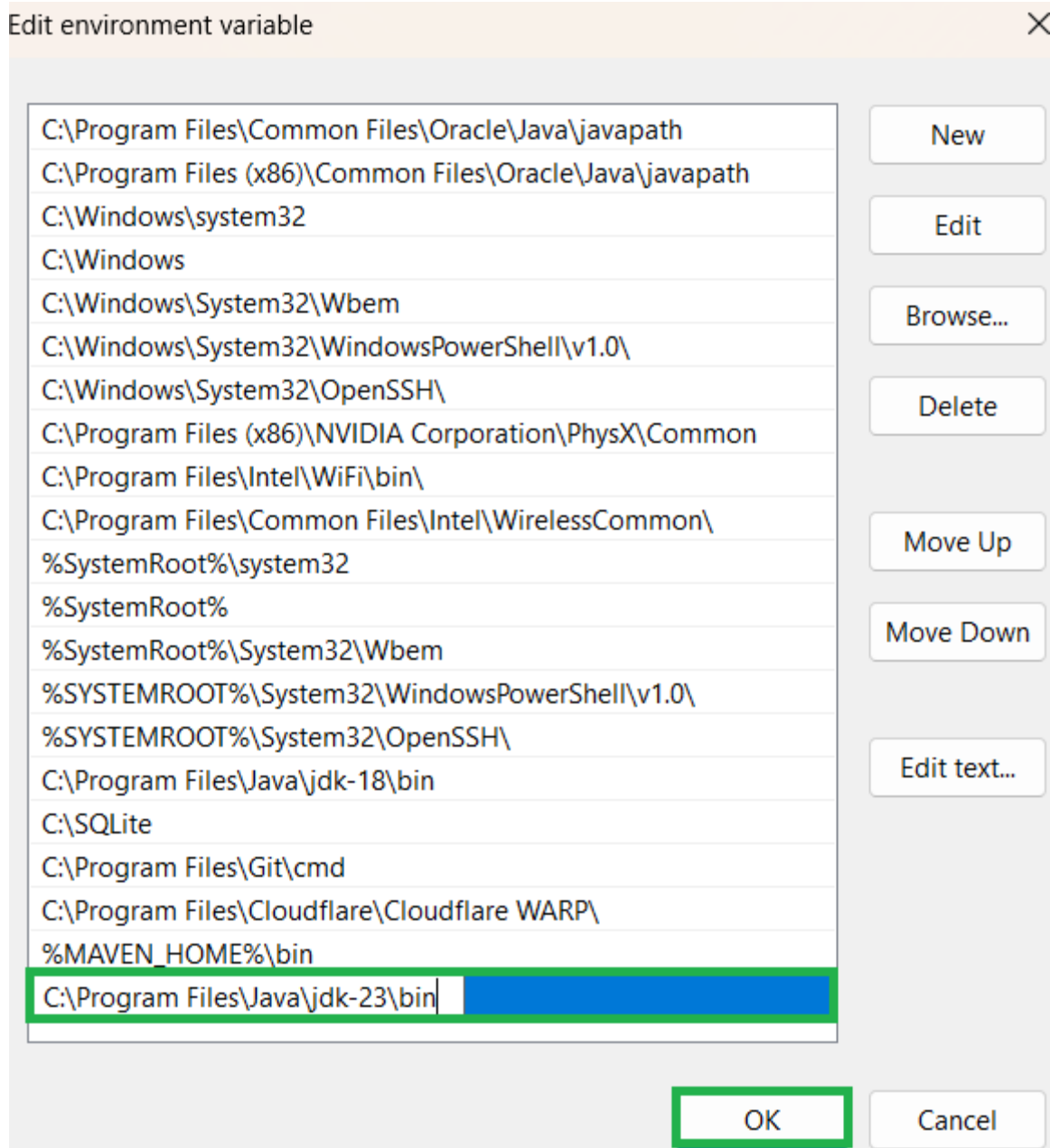
Search for environment variable on your system, then click on the Environment Variables button.



In the system variable section, select the path variable and then click the option Edit.



Paste the path that copied earlier, and **click OK** to save the changes.



Step 5: Open Command Prompt and run `java --version` to verify the Java installation.

```
Command Prompt
Microsoft Windows [Version 10.0.26100.2454]
(c) Microsoft Corporation. All rights reserved.

C:\Users\GFG0574> java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\GFG0574>
```

How to Install Jenkins on Windows :-

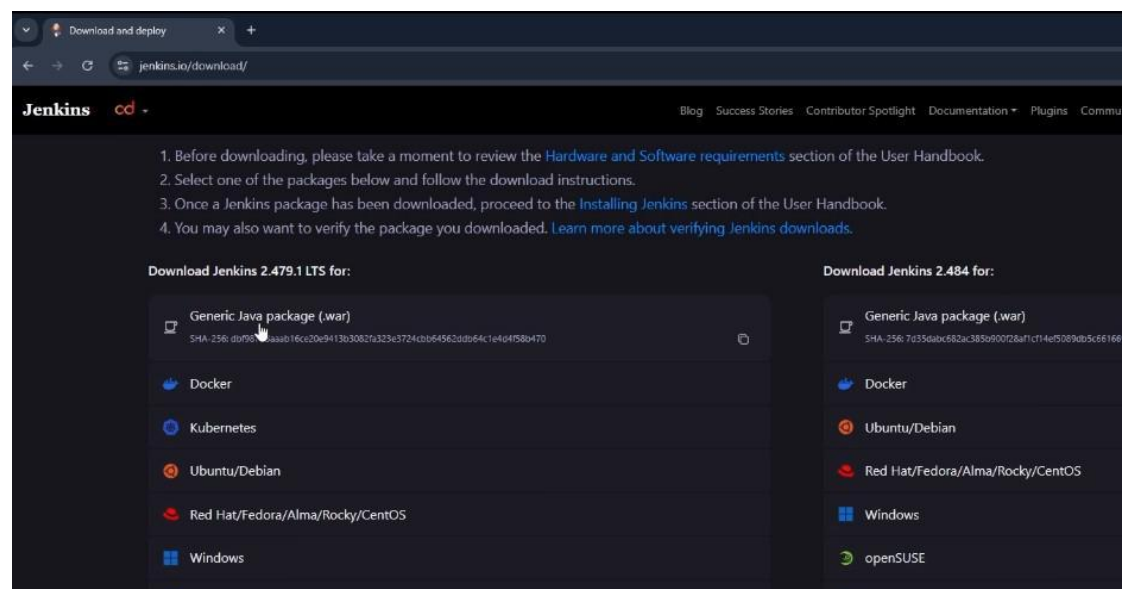
- Jenkins is used for Continuous Integration (CI) and Continuous Delivery (CD).
- It is a Java-based application.
- Jenkins has multiple plugins to automate various tasks at the infrastructure level.
- Plugins provide the different functionalities of Jenkins.

Steps to Installing Jenkins on Windows :-

Here are the step to install Jenkins in your Windows operating system

1. Download the Jenkins WAR File.

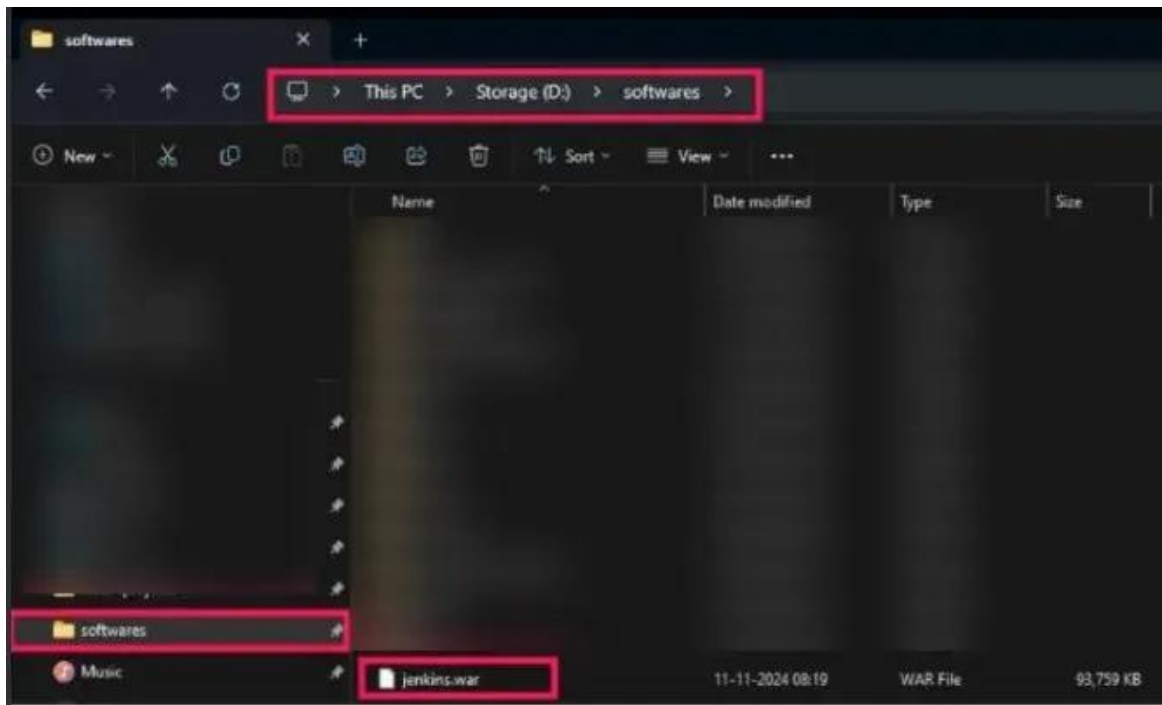
Click on the [Jenkins download](https://jenkins.io/download/) for Windows.



Jenkins Download Page

2. Start Jenkins Using the WAR File.

Place the downloaded jenkins.war file in a folder on your computer to prepare for execution and starting Jenkins.



Place the jenkins war file in a desired folder

Open the terminal or command prompt and navigate to the folder where the jenkins.war file is located.

Run the following command:

java -jar jenkins.war

```

software$ java -jar .\jenkins.war
Running from: D:\softwares\jenkins.war
webroot: C:\Users\DELL\.jenkins\war
2024-11-11 02:55:29.800+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2024-11-11 02:55:33.103+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2024-11-11 02:55:33.238+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.13; built: 2024-09-03T03:04:05.240Z; git: 816018a420329c1caed4116799cda8c60a57cd; jvm 21.0.3+9-LTS
2024-11-11 02:55:34.708+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2024-11-11 02:55:34.809+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2024-11-11 02:55:36.032+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\DELL\.jenkins
Found at: $user.home/.jenkins
2024-11-11 02:55:36.287+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@1431267b{Jenkins v2.479.1,/,b=file:///C:/Users/DELL/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@342726f1{STARTED}}
2024-11-11 02:55:36.321+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@bd1111a{HTTP/1.1,(http/1.1)}{0.0.0.0:8080}
2024-11-11 02:55:36.384+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@40db2a24{STARTING}[12.0.13,sto=0] @7802ms
2024-11-11 02:55:36.397+0000 [id=35] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2024-11-11 02:55:36.850+0000 [id=43] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2024-11-11 02:55:36.884+0000 [id=41] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2024-11-11 02:55:38.845+0000 [id=41] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2024-11-11 02:55:38.868+0000 [id=46] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2024-11-11 02:55:38.871+0000 [id=46] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
  
```

Run the following command

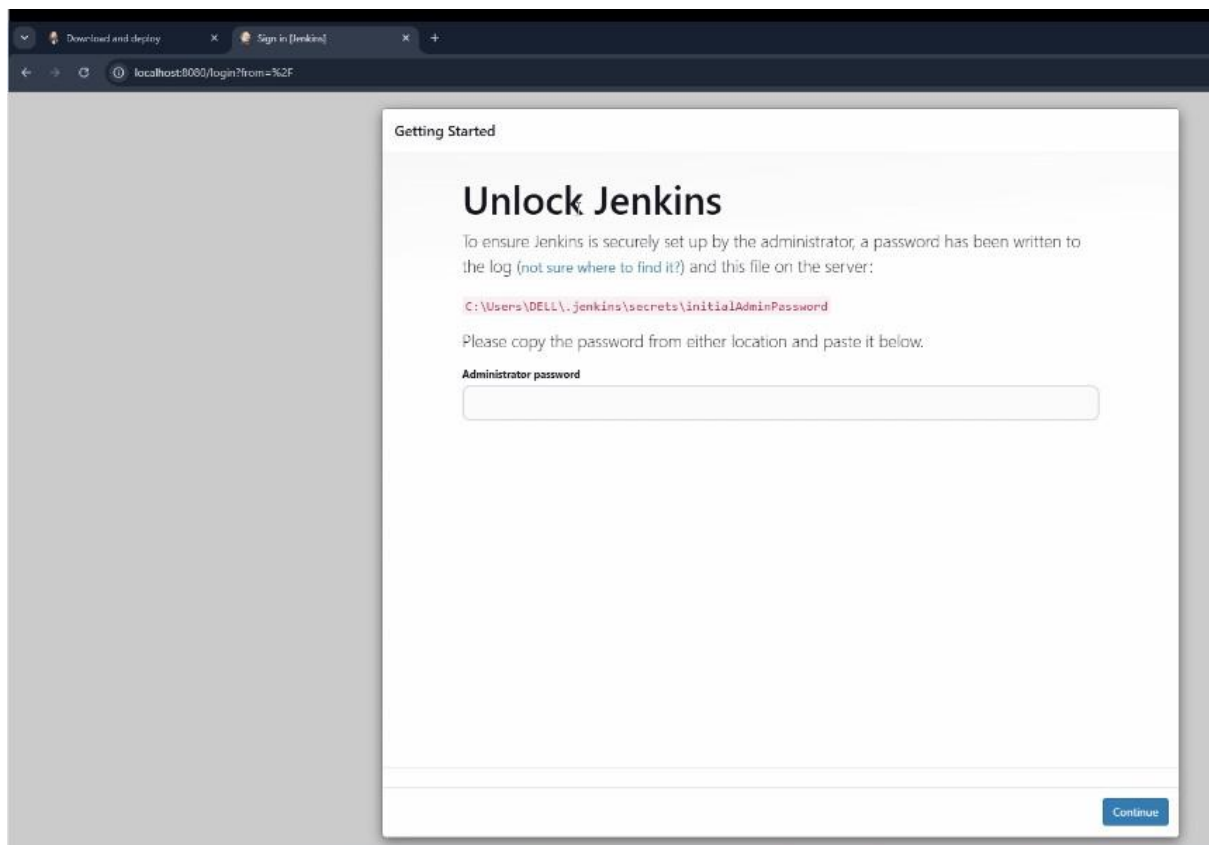
Jenkins will initialize and start up, printing logs to the console. Here are the one time password will be generate to login the System which is :

```
*****
*****
*****
Jenkins initial setup is required. An admin user has been created and a password generated.
Please use the following password to proceed to installation:
10d3965df33e416bb773e50d8db125dc
This may also be found at: C:\Users\DELL\.jenkins\secrets\initialAdminPassword
*****
*****
*****
2024-11-11 02:55:47.773+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2024-11-11 02:55:47.873+0000 [id=34] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
```

3. Access Jenkins on Your Browser.

Open your web browser and navigate to the following URL:

http://localhost:8080



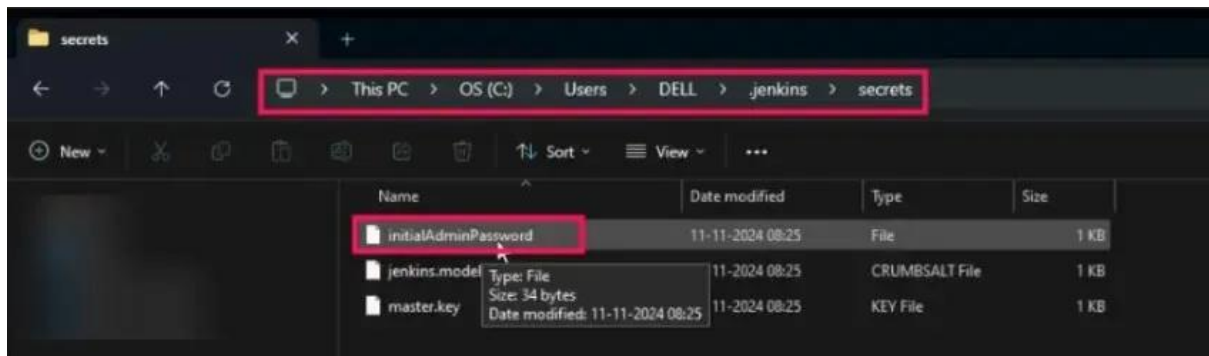
Access Jenkins on Your Browser

By default, Jenkins runs on port 8080. If port 8080 is unavailable, Jenkins will automatically suggest an alternate port in the terminal output.

4. Unlock Jenkins.

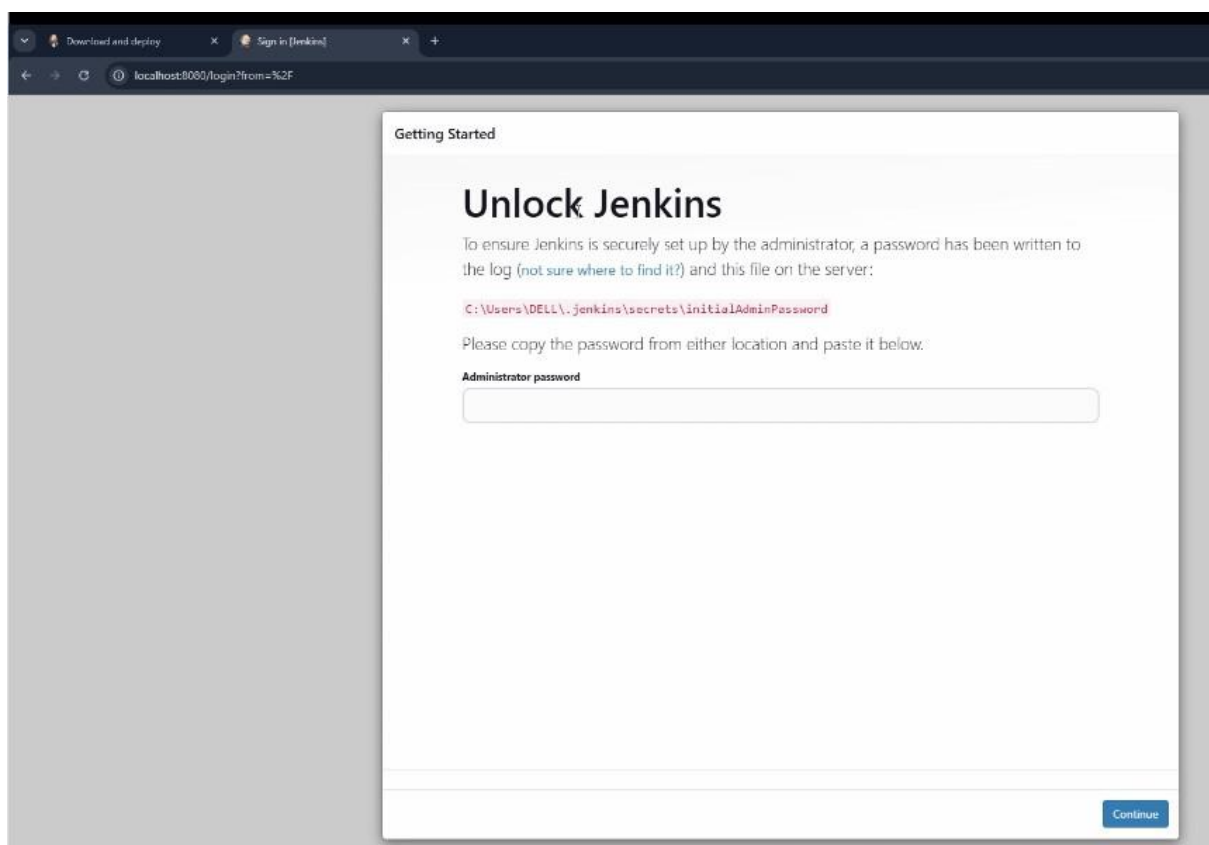
During the setup, Jenkins will generate a one-time admin password. Retrieve the password from the console logs or the initialAdminPassword file located in:

`C:\Users\<YourUsername>\.jenkins\secrets`



initialAdminPassword file located

Enter the password on the Jenkins unlock page to proceed.



Enter the password on the Jenkins unlock page

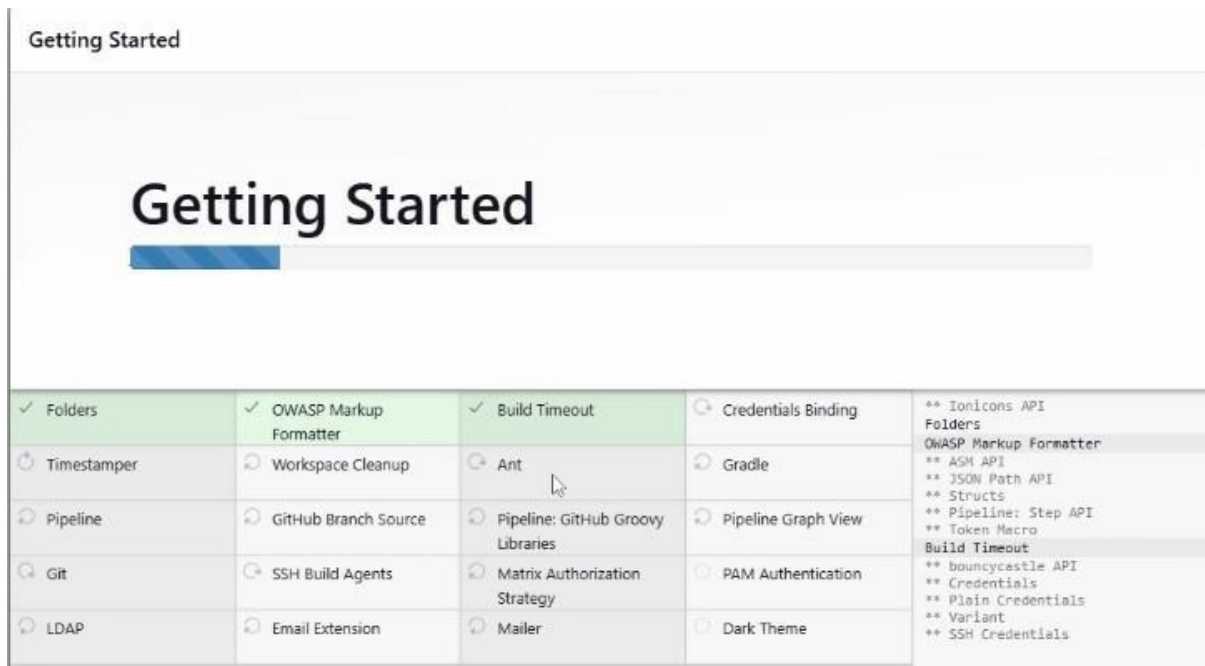
5. Install Suggested Plugins.

Jenkins will now prompt you to install plugins. Plugins in Jenkins provide extended functionality, allowing you to integrate with various tools and platforms.



install plugins

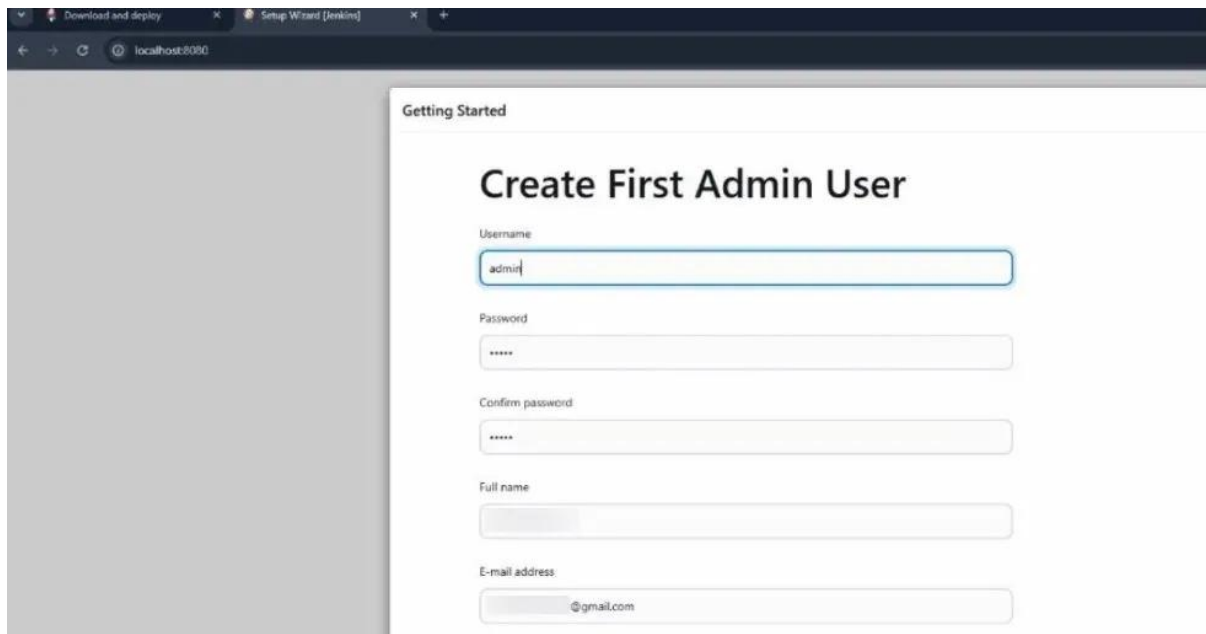
Choose Install Suggested Plugins for a quick setup. This will install commonly used plugins like Git, Pipeline, and Email Notification.



Install commonly used plugins like Git, Pipeline, and Email Notification.

6. Create an Admin User

Set up an admin user with a username, password, and email.

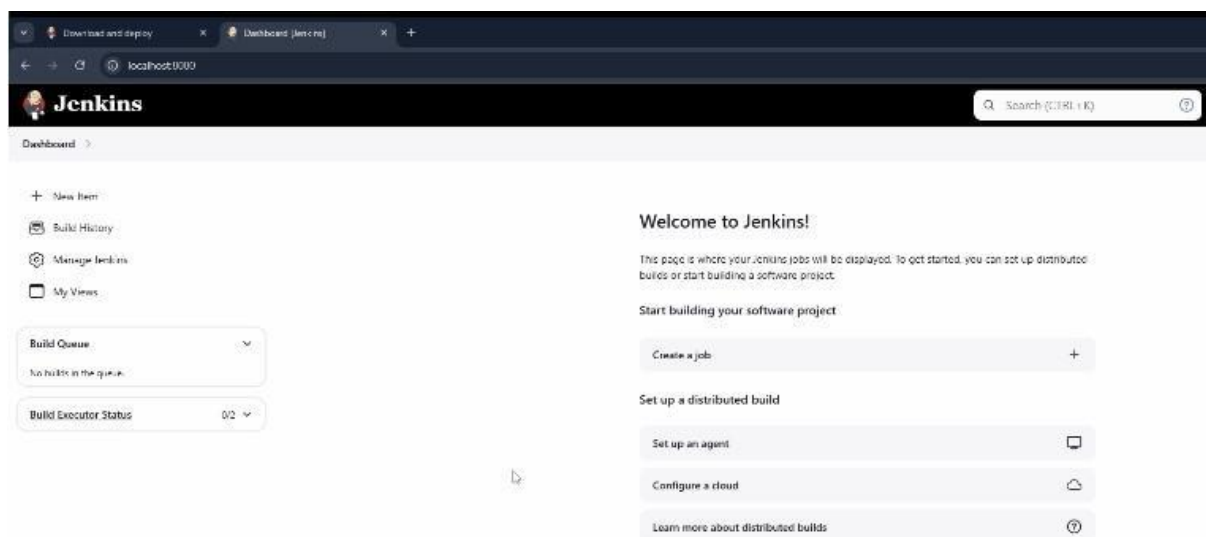
A screenshot of a web browser showing the Jenkins 'Getting Started' page. The main heading is 'Create First Admin User'. Below the heading are five input fields: 'Username' (containing 'admin'), 'Password' (masked with dots), 'Confirm password' (masked with dots), 'Full name' (empty), and 'E-mail address' (containing '@gmail.com'). The browser's address bar shows 'localhost:8080'.

Create an Admin User

This user will be used for future logins.

7. Complete Installation

After setting up your admin user, Jenkins will finalize the installation and redirect you to the Jenkins dashboard.



Dashboard of Jenkins

Here are we successfully installed Jenkins on your Windows machine.